Paleomagnetism of Late Proterozoic rocks, Victoria Island, Northwest Territories, Canada

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Late Proterozoic Natkusiak plateau basalts and associated dolerite dikes and sills of Victoria Island show paleomagnetic polarities that relate them to the Franklin diabases but are complicated by the effects of a downward component that cannot be wholly removed in all sites by AF and thermal demagnetization. In a preserved thickness of about 740 m of Natkusiak flows the polarity changes upward in roughly successive thirds of the section from east and slightly downward (reverse), to west and slightly downward (normal), to east and slightly upward (reverse). Thus there is a double reversal in the section but the lower reverse direction is badly contaminated by a downward directed overprint. At three sites this steep downward component persists after cleaning. Dolerite polarities, except for one that is normal, cluster in the east-down (reverse) direction close to the lower reverse direction of the lavas and steeply downward, similar to the overprint direction in the flows. The downward component can be correlated with evidence of low initial oxidation and more severe alteration in the corresponding samples and may be related to burial metamorphism. The reverse direction of the upper lavas, being from the freshest parts of the section and closest in direction to the exact reversals recorded in Franklin dikes from Baffin Island, is believed to be most reliable. Paleopole positions are: upper reverse lavas, 161°E, 2°N; middle normal lavas, 159°E, 9°N; combined middle–upper lavas, 159°E, 6°N ($\delta p = 4°$, $\delta m = 8°$).

Reynolds Point shales from the Shaler Group underlying Natkusiak basalts and host to the dolerite sills yielded a paleopole position at 147°E, 6°S, significantly removed from the Franklin poles but probably not indicative of an age greater than Hadrynian.

Les basaltes des plateaux de Natkusiak du Protérozoïque supérieur et les dykes et filons-couches associés de dolérite de l'île Victoria présentent des polarités paléomagnétiques qui les apparentent aux diabases de Franklin non pas sans complication due aux effets de la composante vers le bas impossible à éliminer complètement à chaque site par la désaimantation thermique et par champ alternatif. Dans une section bien conservée de coulées de Natkusiak d'une puissance de 740 m la polarité varie vers le haut, approximativement à tous les tiers de la section, de l'est et faiblement vers le bas (inverse) vers l'ouest et faiblement vers le bas (normal) et faiblement vers le haut (inverse). Donc on observe une double inversion dans la section mais la direction inversée à la base est fortement oblitérée par une superposition dirigée vers le bas. Cette composante plongeant abruptement vers le bas résiste au nettoyage à chacun des trois sites. Les polarités des dolérites, à l'exception de celle normale, se regroupent avec une direction est – vers le bas (inverse) se rapprochant de la direction inversée à la base des laves et elles plongent abruptement tout comme la direction dans les coulées affectées par la superposition. La composante vers le bas peut être corrélée avec l'existence d'une faible oxydation et d'une altération plus prononcée des échantillons correspondants et peut être reliée à un métamorphisme d'enfouissement. La direction inversée des laves du haut qui constituent les parties les plus fraîches de la section et dont les directions sont les plus cohérentes avec les inversions exactes enregistrées dans les dykes de Franklin de l'île de Baffin est semble-t-il la direction la plus réelle. Les positions des paléopôles sont: laves du haut de la section avec inversion, 161°E, 2°N; laves du centre avec direction normale, 159°E, 9°N; laves d'en haut et du centre combinées, 159°E, 6°N ($\delta p = 4°$, $\delta m = 8°$).

Les schistes argileux de Reynolds Point du groupe Shaler sous-jacents aux basaltes de Natkusiak et qui forment la roche encaissante des filons-couches de dolérite ont fourni une position de paléopôle à 147°E, 6°S, passablement différente des pôles de Franklin mais qui n'indique pas un âge probablement plus ancien que Hadrynien.

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Introduction

The Natkusiak basalts and accompanying dolerites of Victoria Island, Northwest Territories (Fig. 1) are part of a late Precambrian plateau basaltic province distributed widely across the northern part of the Canadian Shield. Elsewhere it is represented by the Franklin dikes (Fahrig *et al.* 1971) and Coronation sills (Robertson and Baragar 1972) and only on Victoria Island are extrusive equivalents present. The Natkusiak Formation, therefore, serves as the stratigraphic standard for the province. An earlier paleomagnetic and age study of

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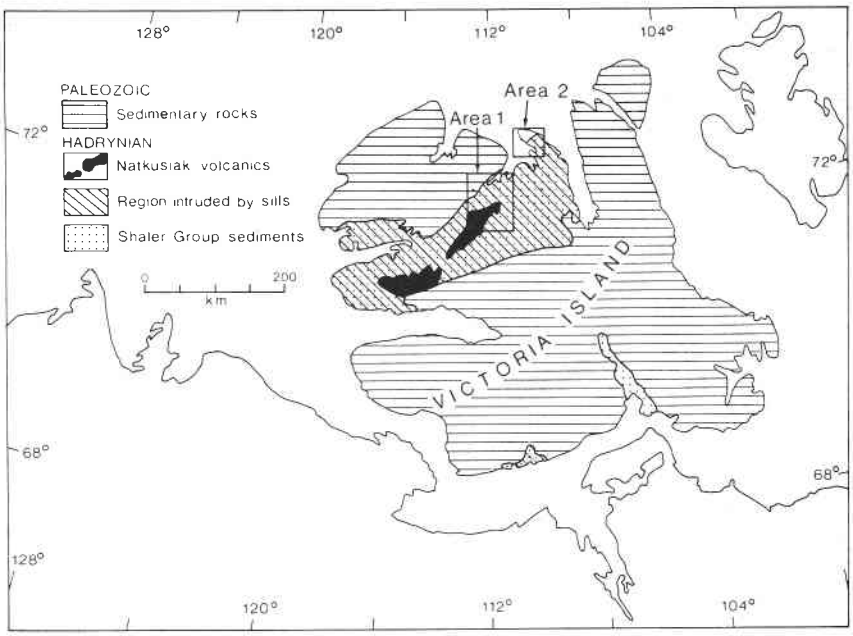


FIG. 1. Generalized geological map of Victoria Island with study areas indicated.

Franklin lavas and dolerites from Victoria Island (Palmer and Hayatsu 1975) was very limited in scope. All samples studied came from two geographically restricted areas and only the lowermost flows of the Natkusiak Formation were examined. In this study the probable maximum exposed thickness of the Natkusiak Formation has been sampled. The dolerites have been sampled over a wider geographic area and over a wider stratigraphic range. In addition the first paleomagnetic results from rocks of the Shaler Group of Victoria Island are also reported.

Paleomagnetic sampling of the basalts and dolerites was done by J.H.F. and M.F. The paleomagnetic samples were measured at the University of Western Ontario by H.C.P. and M.F. Results of the stratigraphic work were published by Baragar (1976), who is now preparing a comprehensive report on the petrochemistry.

General geology

The Natkusiak Formation rests with slight discordance on the Hadrynian Shaler Group of predominantly carbonates and sandstones with minor shale and gypsum (Thorsteinsson and Tozer 1962). The Shaler Group lies unconformably on Archean basement and comprises five formations of which the second oldest, the Reynolds Point, contains shales sampled in the present study. Dolerite sills, assumed to be the precursors of the Natkusiak basalts, intrude the sediments at all stratigraphic levels. These are numbered in succession downward from the base of the Natkusiak Formation to sill No. 13 (Fig. 2), the lowest level sampled for this study. Northwesterly trending dolerite dikes can be observed both feeding and cutting sills and flows to high stratigraphic levels, and hence are interpreted as source conduits active throughout the magmatic episode. All rocks are gently folded into a northeasterly trending syncline with dips generally less than 10° but the Natkusiak Formation is distributed along its axis and dips are essentially flat.

The Natkusiak Formation comprises a succession of flows ranging in thickness from 15 to 60 m and having a maximum preserved aggregate thickness of about 740 m (Fig. 3). Its stratigraphy was measured along several discrete sections and correlated by means of flows that were traceable from section to section (Fig. 3). Flow samples were taken along section lines and are fixed with respect to the stratigraphy.

Age

The late Hadrynian age assumed for the magmatic rocks is based upon a K–Ar age determination of 640 Ma for one of the sills reported by Christie (1964) and a K–Ar isochron age of 625 Ma determined for the basalts by Palmer and Hayatsu (1975). Confirmation of these ages by a Rb–Sr isochron was attempted; the Rb/Sr ratios in the dolerites are uniformly low with limited spread and cluster about a 2.2 Ga reference isochron (Baragar and Loveridge 1982). An age of 2.2 Ga is geologically unreasonable. This result is thought to reflect contamination of late crystallizing fractions of the magma by solutions from Archean basement through which the magma passed. However, subsidiary isochrons based upon basalt samples believed

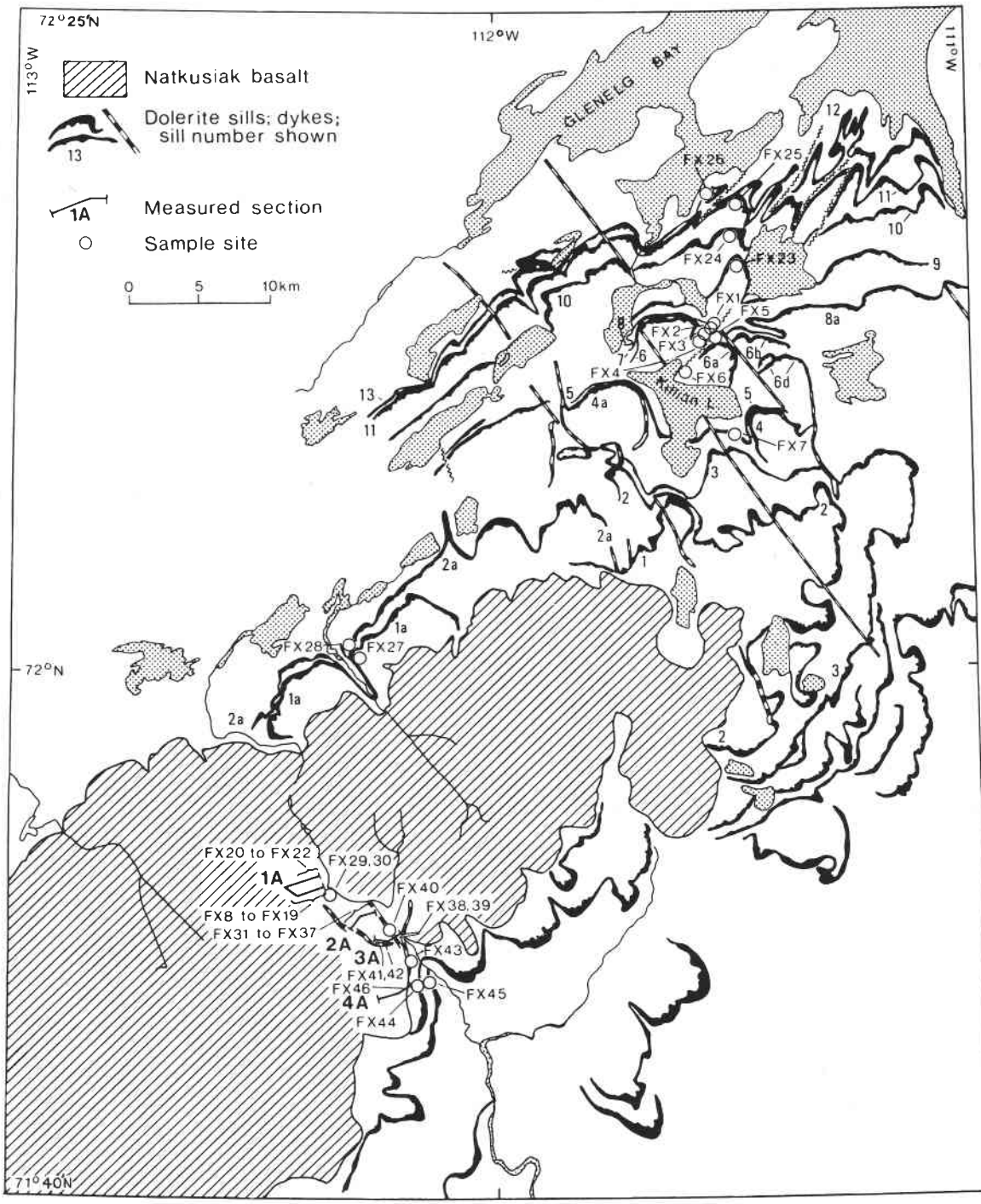


FIG. 2. Igneous rocks of the northeastern part of the Minto Arch, Victoria Island. The sills are numbered from 1 to 13 in descending stratigraphic order from the base of the Natkusiak Formation. Paleomagnetic sampling sites are indicated by open circles and the site numbers are preceded by the letters FX.

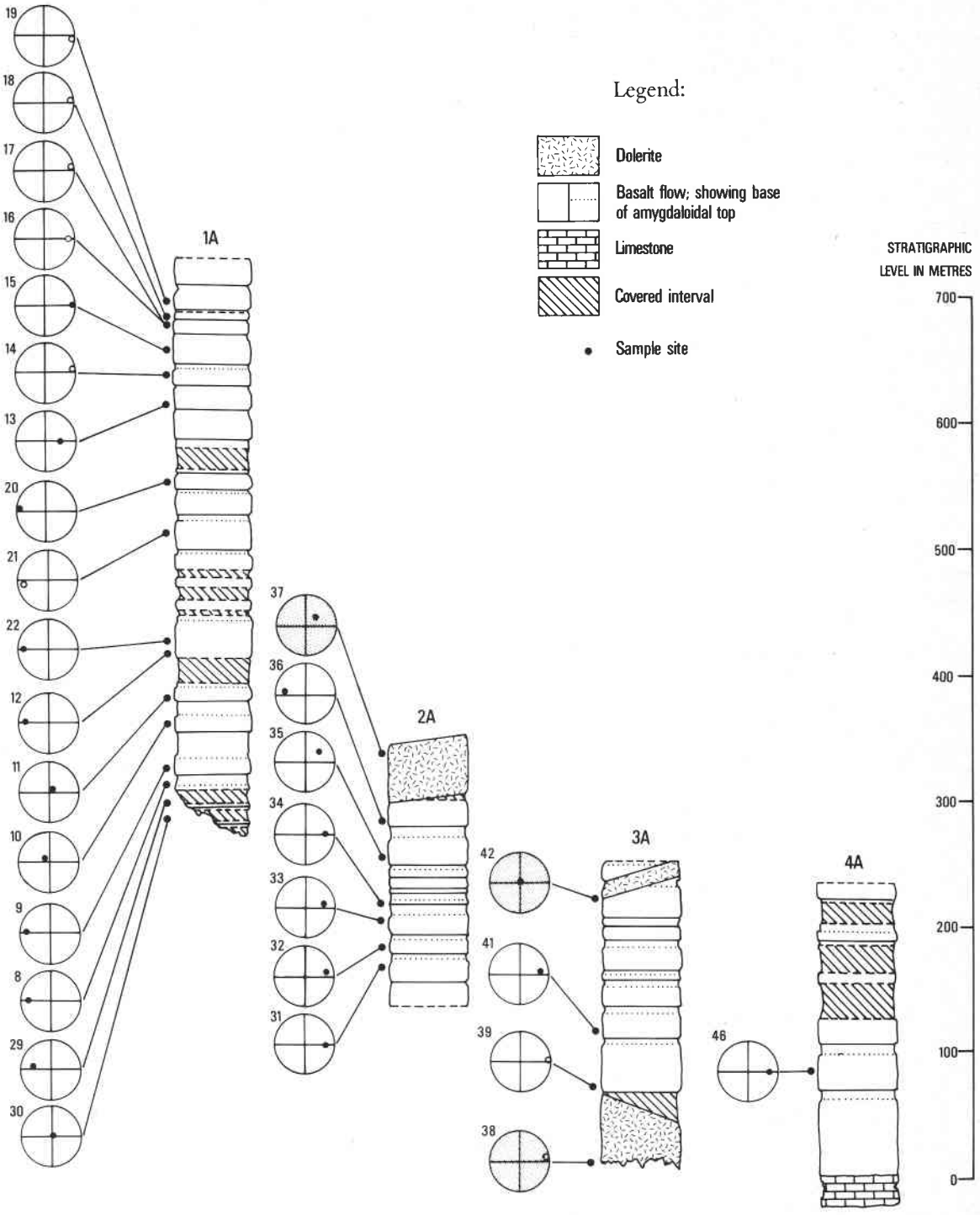


FIG. 3. Measured stratigraphic sections at profiles 1A, 2A, 3A, and 4A of Fig. 2, showing the locations of paleomagnetic sample sites within the Natkusiak Formation and dolerite sills intrusive into it. Small equal-area projections to the left of each column indicate the characteristic magnetization (from Table 2) at each lava site. In these plots solid (open) circles represent projections on the lower (upper) hemisphere. Equal-area projections of the dolerite sites are shaded.

to be little contaminated give ages of 776 and 705 Ma. Although these latter ages cannot be accepted with any degree of confidence, in conjunction with the K–Ar ages they are indicative of a very Late Precambrian age for the Natkusiak igneous event. The age commonly accepted for the Franklin magmatic event is 675 Ma (Fahrig *et al.* 1971).

Paleomagnetic field and laboratory methods

The locations of paleomagnetic sampling sites within the dolerites (24 sites) and the shales of the Reynolds Point Formation (10 sites) are shown in Figs. 2 and 4. Also shown in Fig. 2 are the locations of traverses along which sampling (26 sites) of the basalts of the Natkusiak Formation was carried out. The 26 sampling sites in basalt are fairly evenly distributed over the 740 m of exposed thickness (Fig. 3).

In the field a total of 360 sample cores, generally six per site, was obtained using a portable drill. At 52 of the 60 sites a solar compass was used to obtain the azimuthal component of orientation. Topographic sightings and magnetic compass bearings were used at the remaining eight sites. The cores were sliced into specimens 2.4 cm in height, yielding a total of 885 specimens and material for polished thin sections. Following initial natural remanent magnetization (NRM) measurement on a Schonstedt spinner magnetometer (model DSM-1), alternating field (AF) demagnetization was performed on at least one specimen per core to peak fields of at least 50 mT in 5 mT increments. The increments were increased to 10 mT for specimens that were demagnetized to fields in the 60–100 mT range. Thermal demagnetization (using a Schonstedt TSD-1 unit) was performed on many of the second specimens from the basalt and dolerite cores. In relatively few samples, AF demagnetization of the first specimen was followed by thermal demagnetization. About 6000 determinations of remanence direction and intensity following a demagnetization step were performed.

Reynolds Point Formation

The Reynolds Point Formation consists essentially of limestone and sandstone (Thorsteinsson and Tozer 1962). Dark grey splintery shale forms a minor constituent and this was sampled for paleomagnetic study. All 10 sites (Fig. 4) are from a scarp on the north limb of the Holman Island syncline; the sites span a stratigraphic thickness of 44 m. Because of the fissile nature of the shales, all cores were drilled subparallel to bedding in order to obtain orientable material. Nevertheless all cores required extensive gluing before specimen preparation; this process effectively eliminated the possibility of thermal demagnetization of these samples. Because of this, magnetic stability of the shales was examined solely by AF demagnetization, which was performed on one specimen from each core.

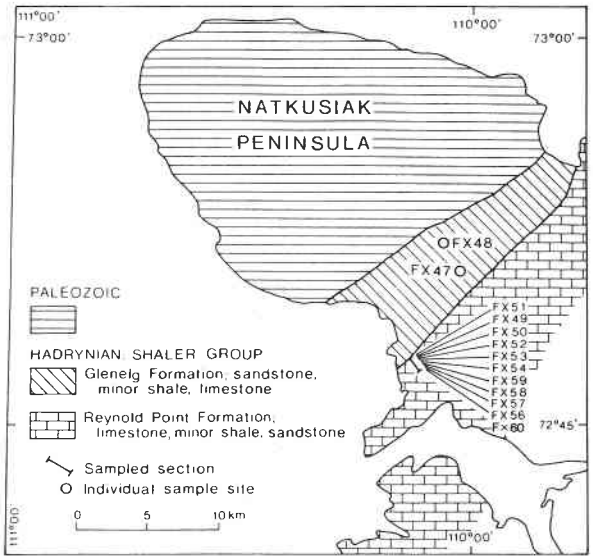


FIG. 4. Geological map of the Natkusiak Peninsula (area 2 of Fig. 1) showing sampling sites in dolerites and in shales of the Reynolds Point Formation.

The shales are weakly magnetized with the NRM in the range $4–32 \times 10^{-4}$ A/m. The NRM directions are moderately well grouped (Fig. 5*a*). Intensity of magnetization is reduced rather rapidly with AF demagnetization (mdf's (median destructive fields) in the range 6–10 mT) but the directions remain coherent. Orthogonal demagnetization diagrams (Fig. 6) suggest that 10 mT is sufficient to isolate the characteristic remanence of these samples. Jagged demagnetization trajectories toward the origin of these plots (Fig. 6) are thought to reflect measurement noise rather than additional components of paleomagnetic significance. In each demagnetization experiment, the last few determinations of remanence required long integration times and approached the effective detection limit of the magnetometer.

Site-mean directions were calculated for each demagnetization step and these were examined for directional end points. Results of this analysis are presented in Fig. 5*b* and in Table 1. The formation mean shifted to a shallower inclination and the precision of the mean improved slightly; k increased from 40 in Fig. 5*a* to 48 in Fig. 5*b*. The Pacific hemisphere pole derived from the structurally corrected result (Table 1) is at 147°E, 06°S ($\delta p = 4°$, $\delta m = 7°$).

Natkusiak basalts

The initial NRM site-mean directions for the 26 lava sites are shown in Fig. 7*a*. The distribution of NRM directions is very similar to that found in Franklin dikes from Baffin Island (Fig. 2 of Fahrig and Schwarz 1973). This pattern suggests that two polarities are represented but that many of the samples have a downward directed overprint.

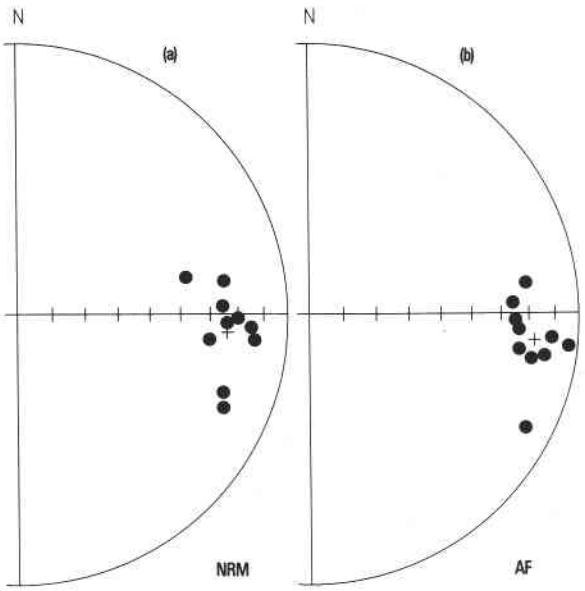


FIG. 5. Structurally uncorrected site-mean directions of magnetization from shales of the Reynolds Point Formation. The closed circles represent north-seeking directions on the lower hemisphere and the cross indicates the mean in each plot. The directions illustrated in (*b*) are those obtained after AF demagnetization and are also recorded in Table 1.

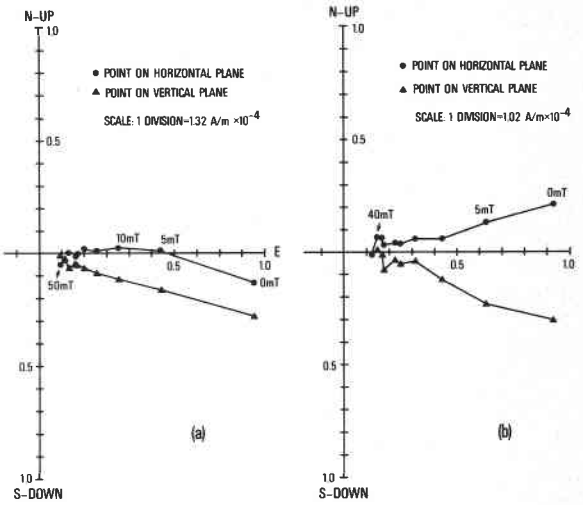


FIG. 6. Orthogonal projection showing successive end points of the magnetization vector after each AF demagnetization step (0–50 mT) for two shale specimens from the Reynolds Point Formation. Peak alternating fields are given beside a few horizontal projections. (*a*) Specimen FX75 53 05A. (*b*) Specimen FX75 56 04A.

AF demagnetization was performed on one specimen from each of the six samples (cores) at each of the 26 lava sites. The site-mean direction that appeared to represent a directional end point for the site was chosen as representative and this generally yielded the highest internal precision. In a few cases, mean directions having lower precision were chosen when there was evidence that systematic directional movement was not complete after cleaning at lower fields. The results obtained from this analysis are shown in Fig. 7*b*. In comparison with Fig. 7*a*, the site means are more tightly grouped about a 180° east–west girdle. Some site means have moved to shallower inclinations; none have moved in the opposite sense, i.e., from shallow to steep inclinations. This is interpreted to mean that both Franklin polarities, as found by Fahrig and Schwarz (1973), are present in the lava data but that in most samples the NRM consists of either an east and shallow or a west and shallow component together with a steep-down component. The coercivity spectra of the steeply down and the shallow components are overlapped to a high degree.

Thermal demagnetization, performed on specimens from 19 of the 26 lava sites, was more effective than AF cleaning in separating the two components in the lava sites. Sites whose characteristic directions were found to be either east and shallow (reverse) or west and shallow (normal) upon AF demagnetization remained as such upon thermal demagnetization. Most sites typified by directions that were either east and steeply inclined downward or west and steeply inclined downward throughout the course of AF demagnetization moved to shallower inclinations over a range of temperatures as illustrated by Figs. 8 and 9.

Final estimates of the characteristic directions were made by two methods. The first method was that outlined above in connection with the AF data in which site means were calculated at each demagnetization level and the "best" direction was chosen based upon evidence of a directional end point of highest internal precision. These directional results, together with the AF field or temperature at which they were obtained, are given in Table 2. The second was a least-squares method based on principal component analysis developed by Kirschvink (1980). With visual reference to an orthogonal demagnetization plot, the orientation of the most representative anchored line was recorded for each lava specimen. Lines having maximum angular deviations (Kirschvink 1980) greater than 5° were excluded. The directions of anchored lines, one per sample, were then averaged by the method of Fisher (1953). The directional estimates found by this method of analysis are also recorded in Table 2.

Table 2 reveals that there is generally good agreement between the two methods. Notable exceptions occur in the data from sites 20, 33, 34, and 35 where the directions obtained from anchored lines are consistently steeper than those found by the more traditional method. These discrepancies arise principally from small but consistent directional shifts that take place during the last few steps of demagnetization. Furthermore the

TABLE 1. Site-mean directional data for Reynolds Point Formation shales

Site	N	Demagnetization level (mT)	$D°$	$I°$	R	k	$\alpha_{95}°$
49	6	15	092	+25	5.731	19	16
50	6	40	101	+17	5.549	11	21
51	6	15	097	+04	5.926	67	8
52	6	15	118	+11	5.966	148	5
53	6	25	100	+13	5.945	90	7
56	6	10	087	+25	5.929	70	8
57	6	10	095	+11	5.616	13	19
58	6	10	100	+22	5.833	30	12
59	6	20	082	+20	5.880	42	10
60	6	10	094	+23	5.844	32	12
Mean	10		097	+17	9.813	48	7
Mean	10	Tilt corrected	101	+18	9.813	48	7

NOTES: N is the number of cores; D and I are the declination and inclination of the mean direction, and R is the resultant; k and α_{95} are Fisher's precision and circle of confidence (P = 0.05) of the directions (Fisher 1953).

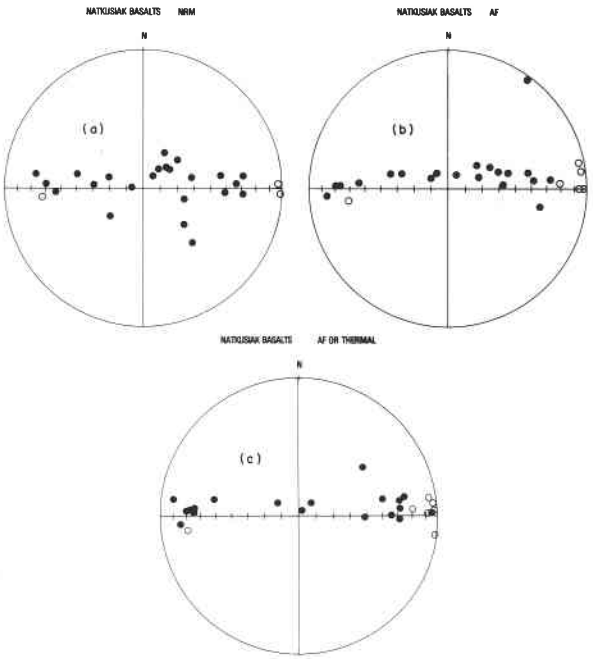


FIG. 7. Lava site means for the Natkusiak basalts (a) as measured initially; (b) after AF demagnetization; and (c) after AF or thermal demagnetization. The results depicted in (c) are means computed from the results of methods 1 and 2 as recorded in Table 2. In all projections, closed (open) circles are projections on the lower (upper) hemisphere.

directional shifts are accompanied by only small-magnitude changes and the absolute magnitudes of these vectors are very small in relation to the magnitudes of the vectors determined at lower temperatures. The highest temperature data points plot very close to the origin and contribute very little weight to the poorly constrained anchored lines, the orientations of which are largely controlled by the lower temperature, higher intensity, steeper vectors. Two examples, from specimens of sites 33 and 35, serve to illustrate this particular problem. These examples are illustrated in Fig. 10. Specimen 35 04B (Fig. 10a) exhibits a pattern that can be described as a curved trajectory to the origin. Specimen 33 03B (Fig. 10b) exhibits a linear segment, formed by the last two data points only, of lesser slope than the linear segment controlled by the lower temperature, higher intensity data points. Components of magnetization represented by two-point lines are not recovered in the method of Kirschvink (1980). This is, in general, a good feature of the method since the reality of distinct components of magnetization is made more secure as the number of data points defining the component increases. Characteristic components of magnetization based on two-vector determinations (or even a single end point) can only be justified if they are found in only a small fraction of a paleomagnetic collection and their acceptance improves overall grouping of site directions.

As can be seen in Fig. 7c, the majority of lava sites are either east and shallow (reverse) or west and shallow (normal) but the implied reversals in polarity are not consistently antiparallel due mainly to a bias present in the lower reverse zone. In applying the first method of analysis there is the possibility of selecting those site-mean directions that tend to reduce the overall asymmetry. In contrast, the use of anchored lines to define characteristic remanence directions tends to enhance the asymmetry of the reversals and this is particularly evident in the analysis of results from the

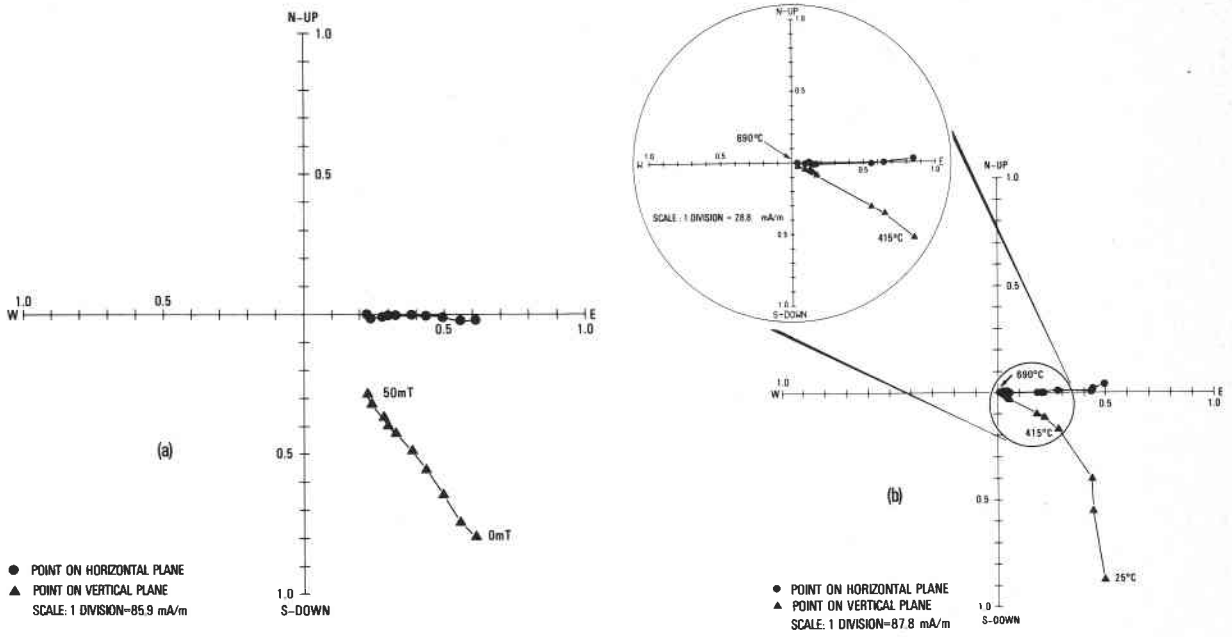


FIG. 8. Orthogonal demagnetization diagrams depicting the contrasting results obtained by AF and thermal demagnetization from two specimens of a single sample. The characteristic magnetization seen by AF demagnetization (*a*) is east and steeply downward. In contrast, thermal demagnetization (*b*) reveals an additional shallower component at temperatures above 415°C. (*a*) Specimen FX75 31 05A, AF treatment (0–50 mT). (*b*) Specimen FX75 31 05B, thermal treatment (25–690°C); inset shows results for specimen FX75 31 05B at temperatures between 415 and 690°C.

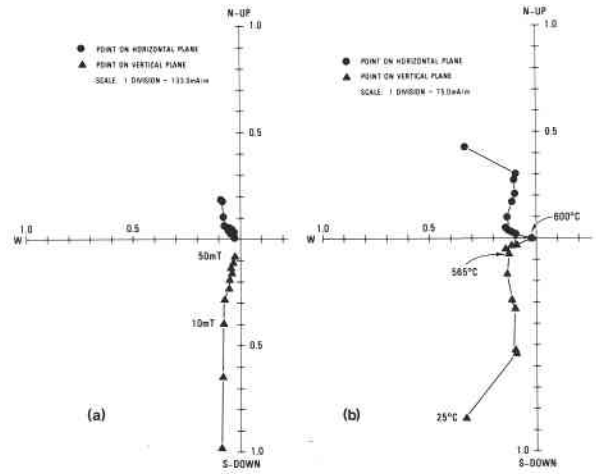


FIG. 9. Orthogonal demagnetization diagrams illustrating the contrasting results obtained by AF and thermal demagnetization from two specimens of a single sample. In contrast with Fig. 8, the AF and low temperature vectors are west and steeply downward. At temperatures above 565°C, however, a westerly directed vector with small positive inclination is revealed. (*a*) Specimen FX75 20 02A, AF treatment (0–50 mT). (*b*) Specimen FX75 20 02B, thermal treatment (25–600°C).

lower reverse zone. In the upper reverse and middle normal zones the two methods yield results that are indistinguishable. The results obtained by method one and method two have been averaged and these average directional estimates are plotted in Fig. 7*c* and recorded in Table 2.

The two major polarity groups of Fig. 7*c* are distributed in an ordered stratigraphic fashion as is illustrated in Fig. 3. The lower flows are characterized by directions that are east and down. The lowest flow sampled (site 39) is an exception; its remanence may have been reset by the underlying sill (site 38). The uppermost flows are also magnetized in an easterly direction but with subhorizontal inclinations. The middle portion of the stratigraphy (from the 280 m level to the 550 m level) is characterized by flows with the westerly directed remanence. Three flows within this normal zone possess steeply down characteristic remanence directions. The generalized polarity pattern over the 700 m sampled section is reverse–normal–reverse. The mean direction of the lower reverse zone (Table 3) is significantly different from the mean directions of the upper reverse zone and the reciprocal polarity of the middle normal zone. This is evidence that the overprint contamination is most pronounced in the lowest part of the volcanic pile. The upper reverse zone is, within the

TABLE 2. Site-mean directional data for Natkusiak basalts

Site	Mean at given demagnetization level, (1)				Mean of anchored lines, (2)			Mean of (1) and (2)	
	Demagnetization level	$D°$	$I°$	(N, k)	$D°$	$I°$	(N, k)	$D°$	$I°$
FX75 19	575°C	098	−01	(6, 219)	098	−01	(5, 216)	098	−01
FX75 18	50 mT	090	−03	(6, 146)	085	−01	(6, 124)	088	−02
FX75 17	50 mT	079	−05	(6, 49)	085	−07	(5, 323)	082	−06
FX75 16	40 mT	087	−21	(7, 122)	087	−18	(6, 276)	087	−20
FX75 15	540°C	086	+02	(6, 26)	091	+06	(4, 38)	089	+04
FX75 14	50 mT	082	−04	(6, 45)	087	−02	(6, 62)	085	−03
FX75 13	515°C	094	+55	(6, 59)	088	+46	(4, 16)	091	+51
FX75 20	595°C	273	+03	(6, 47)	282	+16	(4, 48)	277	+10
FX75 21	625°C	261	−20	(6, 851)	264	−21	(5, 352)	263	−21
FX75 22	50 mT	266	+14	(6, 292)	266	+17	(6, 85)	266	+16
FX75 12	30 mT	272	+20	(6, 36)	273	+20	(3, 58)	273	+20
FX75 11[a]	50 mT	036	+03	(6, 5)	078	+75	(5, 2)		
FX75 11[a]	540°C	046	+78	(6, 32)	037	+81	(4, 28)	042	+80
FX75 10[a]	30 mT	324	+79	(6, 45)	334	+77	(4, 52)	329	+78
FX75 10[a]	590°C	258	+68	(6, 16)	350	+70	(5, 70)	301	+75
FX75 9	15 mT	272	+23	(5, 35)	274	+21	(5, 34)	273	+22
FX75 8	600°C	272	+27	(6, 59)	272	+24	(4, 65)	272	+26
FX75 29	540°C	283	+38	(6, 292)	279	+38	(4, 311)	281	+38
FX75 30	560°C	245	+82	(7, 5)	046	+75	(9, 57)	026	+86
FX75 36	630°C	277	+25	(6, 165)	271	+26	(6, 270)	274	+26
FX75 35	600°C	060	+27	(6, 61)	041	+55	(6, 12)	053	+41
FX75 34	675°C	098	+30	(6, 14)	081	+35	(6, 42)	090	+33
FX75 33	630°C	075	+34	(6, 67)	074	+42	(6, 55)	075	+38
FX75 32	625°C	079	+29	(6, 246)	080	+19	(4, 122)	080	+24
FX75 31	575°C	091	+28	(6, 612)	092	+28	(6, 473)	092	+28
FX75 41	575°C	081	+26	(6, 274)	081	+29	(5, 676)	081	+28
FX75 46	635°C	086	+29	(6, 418)	086	+26	(6, 412)	086	+28
FX75 39	540°C	089	−08	(6, 1414)	089	−08	(5, 286)	089	−08
FX75 37[b]	535°C	053	+54	(6, 91)	048	+59	(6, 43)	051	+57
FX75 42[b]	25 mT	072	+01	(5, 13)	075	−02	(5, 14)	074	−00
FX75 38[b]	20 mT	080	−09	(6, 287)	080	−05	(6, 427)	080	−08

NOTES: (1) This is the mean where the scatter of individual data points (N) was a minimum after the mean direction stopped changing significantly. (2) This is the vector unit mean of all characteristic magnetizations (N) found to decay linearly to the origin of an orthogonal demagnetization diagram after the method of Kirschvink (1980). D and I are the declination and inclination of the mean direction; k is Fisher's precision parameter (Fisher 1953).

[a]Both AF and thermal results are given to illustrate that steep-down vectors are obtained by either method of demagnetization.

[b]Dolerites within the flow sequence.

95% confidence limit, antiparallel to the middle normal zone. The 14 site directions from these two upper polarity zones yield a pole at 159°E, 6°N ($\delta p = 4°$, $\delta m = 8°$).

Franklin dolerites

The dispersal of the NRM site-mean directions (Fig. 11*a*) ranges from steeply downward to easterly and shallow. One site only (site 6 from sill No. 6a) appears to possess normal polarity.

Site-mean directions, resulting from AF demagnetization of the 24 dolerite sites and calculated by method 1, are shown in Fig. 11*b* and recorded in Table 4. By comparing the data in Fig. 11*a* and in Fig. 11*b*, it can be seen that the AF demagnetization has served to make the pattern of directions more bimodal. The majority of directions are either easterly directed with shallow inclinations or are steeply inclined downward. Thermal demagnetization was performed on specimens from 12 of the 24 dolerite sites. The directional results from the thermal demagnetization experiments are also recorded in Table 4 and are illustrated in Fig. 11*b*.

Directional data from the dolerites contrast rather markedly with the directional data from the Natkusiak basalts. (1) Only one sill appears to retain the normal polarity. (2) Thermal demagnetization is much less

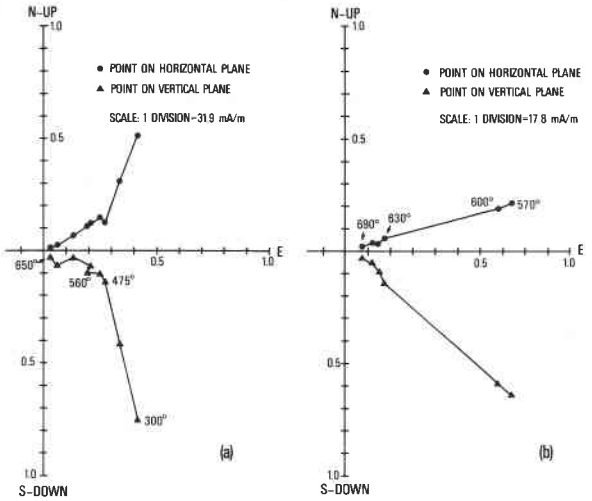


FIG. 10. Orthogonal demagnetization diagrams from two easterly magnetized samples where the final east and shallow component is less well defined than that depicted in Fig. 8. In order to amplify the vector information near the origin, the low-temperature data points have been omitted. In (*a*) the vectors measured above 540°C and in (*b*) the vectors measured above 650°C do not satisfy minimum linearity criteria. In both these examples, poorly constrained, long anchored lines were found but the orientation of these lines is largely controlled by high-magnitude, low-temperature vectors beyond the limits of each of the diagrams. (*a*) Specimen FX75 35 04B, thermal treatment (300–650°C). (*b*) Specimen FX75 33 03B, thermal treatment (570–690°C).

effective in producing coherent directional groupings; in the lava samples, thermal demagnetization preferentially removes the steep-down vector and in most cases reveals the characteristic normal or reverse polarity directions. By contrast, in the case of the sill samples, inclinations after thermal cleaning may be either steeper or shallower than that obtained by AF demagnetization (Fig. 11*b*). (3) The characteristic remanence in the dolerites is not as well defined as in the lavas. This is indicated by the results in Table 4 where the mean directions obtained by the second method of directional analysis are presented. In most sites, fewer than the six measured specimens produced successive vectors that formed linear arrays decaying to the origin of orthogonal demagnetization plots. This latter characteristic is illustrated by the low number of anchored lines found at each site (Table 4). (4) Finally, the steep-down and the reverse polarity (shallow-east) remanence directions cannot *both* be primary remanence directions. Sites 27 and 43 are from sill No. 1 and sites 28 and 44 are from sill No. 2 but in each sill one site yielded the steep component and the other an easterly directed (reverse) component.

Because of the non-Fisherian distribution of the site means contributing to the reverse polarity grouping (Fig. 11*b*), modal analysis (Van Alstine 1980) was employed using the specimen data as the basis for analysis. Both AF and thermal results were used; the directions used from each specimen were those found as lines by the method of Kirschvink (1980). All anchored lines found by this method were accepted together with all intermediate or initial linear segments (unanchored lines) provided that the unanchored lines could be confirmed by vector subtraction. The vector subtraction confirmation technique consisted of first computing vector differences from consecutive measured vectors, then computing vector differences from the first and third measurements, the second and fourth measurements, etc. Four such levels of vector differences were computed from the data of each specimen and these vector differences were machine contoured. Modes on the contour plots were then cross-checked against lines found by the principal component analysis.

By the above techniques, 308 directions were obtained from the demagnetization data of the dolerite specimens. Of these, 86% contributed to two directional modes centered at 024°, +82° and 079°, +12.5°. Subsets of these modes having Fisherian distributions have the following statistics: $N = 129$, $D = 034°$, $I = +82°$, $\beta_{95} = 2°$ and $N = 136$, $D = 079°$, $I = +12.5°$, $\beta_{95} = 4°$. Corresponding paleomagnetic poles are at 51°W, 80°N and 166°E, 09°S.

Discussion of paleomagnetic results

Reynolds Point Formation

The pole position (147°E, 06°S) derived from the shales of the Reynolds Point Formation plots within the region occupied by many Grenville Province poles. Although it is difficult to date the time of remanence acquisition in high-grade plutonic–metamorphic terranes, recent reviews (Roy and Robertson 1978; Irving 1979) place the relevant Grenville poles in the 1000–950 Ma interval or early Hadrynian. Assignment of the Reynolds Point Formation to this age interval is in agreement with the age assignment of the entire Shaler Group (1.1–0.7 Ga) based on large-scale regional stratigraphic considerations (Young 1979).

Natkusiak basalts and Franklin dolerites

Ignoring for the moment data from the dolerites it could be argued that all characteristic remanences found in the lavas as depicted in Fig. 3 and recorded in Table 2 are primary. Under this hypothesis the paleofield would have been eastward and down (reverse), then westward and down (normal), and finally eastward and shallow up (reverse). Sites within the middle normal division possessing steep downward directed remanences (sites 30, 10, 11, and 13) would then be interpreted as having recorded transitional fields near reversal boundaries and (or) abortive reversals.

TABLE 3. Polarity zone directional statistics

Polarity zone	Sites	Method of analysis	N	R	k	$\alpha_{95}°$	$D°$	$I°$	Lat.°	Long.°	δp°	δm°
Upper reverse	19, 18, 17, 16 15, 14	1	6	5.917	61	8	087	−05				
Upper reverse	19, 18, 17, 16 15, 14	2	6	5.932	73	8	089	−04				
Upper reverse	19, 18, 17, 16 15, 14	1 & 2	6	5.929	70	8	088	−05	+02	+161	4	8
Middle normal	20, 21, 22, 12 9, 8, 29, 36	1	8	7.631	19	13	272	+17				
Middle normal	20, 21, 22, 12 9, 8, 29, 36	2	8	7.665	21	12	273	+18				
Middle normal	20, 21, 22, 12 9, 8, 29, 36	1 & 2	8	7.659	21	13	272	+17	+09	+159	6	13
Lower reverse	35, 34, 33, 32 31, 41, 46	1	7	6.890	55	8	081	+29				
Lower reverse	35, 34, 33, 32 31, 41, 46	2	7	6.742	23	13	079	+34				
Lower reverse	35, 34, 33, 32 31, 41, 46	1 & 2	7	6.859	43	9	080	+32	−20	+162	6	10
Steep	11, 10, 30	1	3	2.916	24	26	271	+83				
Steep	11, 10, 30	2	3	2.972	71	15	019	+77				
Steep	11, 10, 30	1 & 2	3	2.966	60	16	348	+83	+85	−143	31	31

NOTES: In the column headed "Method of analysis," 1 and 2 have the meanings given in Table 2, and the entries 1 & 2 are regarded as the best overall summaries; N is the number of sites; R is the resultant of the mean direction given in terms of declination (D) and inclination (I); k and α_{95} are Fisher's precision and circle of confidence ($P = 0.05$) of the directions; Lat. and Long. are the coordinates of the pole with easterly longitudes and northerly latitudes given as +; δp and δm are the semi-angles of the oval of confidence about the pole position.

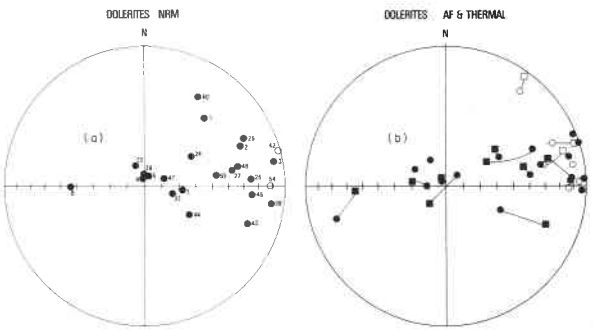


FIG. 11. Site-mean directions for sites in the dolerites as measured initially in (*a*). The NRM directions are identified by sampling site numbers. Circles in (*b*) represent directions after AF demagnetization, whereas squares represent directions after thermal demagnetization. Lines joining circles and squares indicate AF and thermal results from the same site. In both (*a*) and (*b*) closed (open) symbols represent directions on the lower (upper) hemisphere.

This hypothesis can be rejected for two reasons. Firstly, there is lack of internal consistency of remanence within unit (sites 30 and 36; Fig. 3, Table 2). This lack of internal consistency was previously noted in the dolerites as well. Secondly, it can be rejected on mineralogical grounds based on the examination of one polished thin section from each of the lava sites. All flows are dark grey rocks devoid of fine-grained earthy hematite. The principal minerals are augite, plagioclase, and iron–titanium oxides. Plagioclase occurs both as microphenocrysts and as a groundmass mineral; augite is only sparsely phenocrystic. Olivine or pseudomorphs of olivine occur in half the flows as microphenocrysts but never in amounts exceeding 3%. Among the secondary minerals, chlorite is most abundant, occurring interstitially, as amygdule fillings, and as a replacement of olivine. Pumpellyite and, to a lesser extent, prehnite make their first appearance at the 600 m level and occur sporadically downwards in the flow succession.

The preservation of primary remanence with long-term stability in mafic igneous rocks is dependent on the presence of one of three morphological–mineralogical types of iron–titanium oxides. Type 1 consists of small (extending below the limits of optical resolution) particles of magnetite exsolved from high-temperature silicate minerals (Hargraves and Young 1969). Type 2 consists of accessory titanomagnetites subdivided by

TABLE 4. Site-mean directional data for dolerites

Site	Mean at given demagnetization level, (1)				Mean of anchored lines, (2)		
	Demagnetization level	$D°$	$I°$	(N, k)	$D°$	$I°$	(N, k)
FX75 01	15 mT	038	−15	(6, 122)	037	−15	(1, —)
	550°C	035	−04	(6, 32)	035	−03	(3, 66)
FX75 02	25 mT	067	+03	(6, 29)	068	+09	(4, 14)
FX75 03	30 mT	077	−30	(6, 223)	070	−27	(4, 190)
	520°C	074	−15	(6, 279)	072	−20	(1, —)
FX75 04	15 mT	324	+86	(6, 555)	330	+86	(5, 339)
	530°C	332	+84	(6, 518)	326	+84	(6, 447)
FX75 05	25 mT	066	+33	(6, 16)	112	+24	(2, 1)
	570°C	059	+62	(6, 44)	041	+75	(5, 108)
FX75 06	25 mT	253	+20	(6, 100)			
	550°C	267	+36	(6, 209)	268	+37	(5, 169)
FX75 07	20 mT	068	−19	(6, 820)	066	−17	(4, 999)
	550°C	072	−04	(6, 122)	072	−05	(4, 68)
FX75 23	15 mT	299	+69	(6, 55)	353	+75	(2, 92)
FX75 24	20 mT	086	+12	(6, 136)	083	+15	(3, 295)
	500°C	075	+26	(6, 156)	079	+11	(6, 54)
FX75 25	25 mT	092	+04	(6, 15)	092	+12	(5, 14)
FX75 26	20 mT	076	+11	(6, 146)	075	+13	(6, 735)
FX75 27	20 mT	088	−06	(6, 103)	076	−02	(2, 36)
FX75 28	35 mT	045	+81	(6, 2005)	012	+83	(6, 661)
	540°C	223	+76	(5, 64)	019	+87	(5, 50)
FX75 37	25 mT	061	+55	(6, 331)	051	+65	(6, 48)
	535°C	053	+54	(6, 91)	048	+59	(6, 43)
FX75 38	20 mT	080	−09	(6, 287)	080	−05	(6, 427)
FX75 40	10 mT	329	+72	(6, 179)	333	+72	(6, 156)
FX75 42	25 mT	072	+01	(5, 12)	075	−02	(5, 14)
FX75 43	10 mT	272	+79	(6, 159)	273	+83	(5, 105)
	500°C	279	+70	(6, 49)	254	+49	(2, 149)
FX75 44	40 mT	114	+55	(6, 312)			
	575°C	111	+25	(6, 13)	111	+42	(5, 24)
FX75 45	20 mT	087	+12	(6, 547)	088	+14	(6, 396)
FX75 47	60 mT	077	+32	(6, 31)	067	+55	(5, 16)
FX75 48	20 mT	087	+03	(5, 237)			
FX75 54	30 mT	091	−14	(6, 478)	090	−13	(6, 455)
FX75 55	15 mT	082	+39	(6, 194)	083	+40	(4, 129)
	575°C	075	+42	(6, 466)	075	+43	(5, 225)

NOTE: Symbol conventions as for Table 2.

abundant and closely spaced trellis ilmenite lamellae producing small magnetite regions separated by nonmagnetic barriers (Watkins and Haggerty 1965, 1967; Wilson *et al.* 1968; Strangway *et al.* 1968). These titanomagnetites would be in oxidation class C-3 as defined by Haggerty (1976). Type 3 consists of titanohematite and hematite resulting from oxidation to stages higher than that represented by type 2 (oxidation classes C-4 to C-6 of Haggerty 1976).

The original titanomagnetites in the Natkusiak basalts have undergone high-temperature oxidation to varying degrees, with oxidation classes C-1 to C-5 all repre-

sented. Separate ilmenite or ilmenite in magnetite–ilmenite composite grains is present in oxidation states up to the R-5 stage (rutile + titanohematite). Titanomagnetites in oxidation states C-2 and C-3 commonly have the trellis ilmenite replaced by black sphene, a feature characteristic of burial metamorphism (Ade-Hall *et al.* 1971). However, the distribution of these oxidation classes within individual flows is not this broad. Flows whose initial remanence is either clearly normal or reverse contain oxides at the high end of the oxidation spectrum (types 2 and 3), whereas flows whose initial remanence is steep-down and that remains so upon either AF or thermal demagnetization are dominated by magnetites lacking or poor in exsolved ilmenite or its decomposition products. Flows possessing the steep-down characteristic remanence thermally unblock at temperatures below 570°C and lack hematite.

The time of acquisition of the steep-down component is problematical. Because most of the sites possessing the steep-down component as the characteristic remanence occur at or near reversal boundaries it could be argued that these remanences were formed by partial baking of normal flows by reverse flows and vice versa. In this model, the steep-down component would be a vector resultant of two opposed or nearly opposed vectors. This would result in very low resultant intensities of the steep vectors. However, the mean NRM intensity of the samples from the steep-down lava sites is 1.08 ± 0.23 A/m, whereas the mean intensity from the samples yielding normal and reverse polarities is 1.09 ± 0.11 A/m, where the errors quoted are the standard errors of the mean. Thus there is no difference in intensity of the vectors, and the vector addition model can be ruled out. Furthermore, this mechanism could not be applied to the dolerite sills that are separated by thick successions of intervening sedimentary strata.

In most previous paleomagnetic studies of Franklin rocks (Robertson and Baragar 1972; Fahrig and Schwarz 1973; Palmer and Hayatsu 1975; Park 1981) a steep-down component of remanence is observed but in most cases this component could be removed by modest AF demagnetization. This component of remanence was ascribed to viscous acquisition in the present or Bruhnes field. However, in some of the Franklin rocks of the present study, the steep-down component completely dominates the entire coercivity and blocking temperature spectra. It is hard to envisage a mechanism whereby viscous remanence, carried by magnetite and not hematite, acquired at ambient temperatures (and for these rocks much of the annual cycle is at subzero temperatures) could completely replace an older thermal or thermo-chemical remanent magnetization.

A steep-down component of magnetization that resisted AF demagnetization was found in some of the dikes studied by Fahrig *et al.* (1971) from Baffin Island. Because of the spatial proximity of these dikes (group B dikes of Fahrig *et al.* 1971) to the Davis Strait, the remagnetization was ascribed to mild reheating during the early Tertiary opening of the Davis Strait. The most accurate directional estimate of the steep component in the present study is thought to be that derived from the flows (Table 3). The directional estimate from the dolerites (034°, +82°) may well have an easterly bias since it is derived by vector subtraction methods from rocks that have an underlying shallow-east component. The flow pole position (85°N, 143°W) is close to that expected for the Eocene of North America (Irving and Irving 1982). We cannot prove but suggest that the steep-down component is a viscous partial thermoremanent magnetization (VPTRM) blocked in following uplift, erosion, and supracrustal cooling. A consequence of this uplift and erosion may well be the widespread Eureka Sound Formation (Tozer 1968, p. 584; Miall 1975). The uplift itself may be a distal expression of the Eurekan orogeny (Tozer 1968, p. 585) and the Eurekan rifting episode (Kerr 1977).

The earlier studies of Franklin rocks from the western Arctic have yielded single-polarity (reverse) remanence whereas Franklin rocks from the central and eastern Arctic have yielded normal and reverse directions that are statistically antiparallel (Fahrig and Schwarz 1973). Furthermore the derived paleopoles from eastern Arctic sites show a small but significant difference from those derived from western Arctic sites (Palmer and Hayatsu 1975). Recent paleomagnetic work on Franklin diabases from the Brock Inlier of the western Arctic (Park 1981) has reconfirmed this regional difference. Palmer and Hayatsu (1975) suggested that this difference in pole position reflected a real age difference in the time of igneous activity in the western and eastern Arctic. This proposition presupposed that primary remanences uncontaminated by any secondary overprint were obtained in the early studies of the western Arctic samples. Because both polarities have now been obtained from the Natkusiak basalts but, unlike the eastern Arctic dikes, the polarities are not consistently antiparallel, it seems equally plausible that the difference in pole positions is related to a bias in the final directions caused by the steep-down overprint. Although data of the type illustrated in Figs. 8*b* and 9*b* are signatures of the removal of a single component of magnetization, this signature is not absolute proof of the removal of a single component. It may equally well be the signature of the removal of two components in constant proportions.

Although the directions obtained from the lower polarity zone of the Natkusiak basalts are probably slightly biassed by the unremoved steep component, the overall patterns of Natkusiak and dolerite poles have significance in terms of regional stratigraphy. The mean direction from the dolerite sills intrusive into the Shaler

Group is close to that obtained from the lower reverse polarity division of the Natkusiak Formation. This could be interpreted as indicating that the sills were emplaced immediately prior to or contemporaneously with the lower 275–300 m of flows.

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